

NOVOGENIA · AKADEMIE

FREQUENTLY ASKED QUESTIONS BIOLOGICAL AGE

Answers to typical client questions about the analysis of biological age.

INHALT

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This FAQ collection answers the most common client questions about the biological age analysis — how telomeres, oxidative stress and detoxification shape the aging process, and which anti-aging recommendations follow from the genetic profile.

BASICS — TELOMERES & AGING

What are telomeres and why are they important for aging?

Telomeres are protective caps at the end of each chromosome — they contain no genes themselves, but protect genes from being lost during cell divisions. With each cell division, a small piece of telomere breaks off. After 50-70 divisions, they are used up; the cell goes into a sleep mode (senescence) and accumulates as an "old cell" in the body.

What is the difference between chronological and biological age?

Chronological age is your life time in years — fixed and unchangeable. Biological age shows how "old" your cells actually are. Two people with the same chronological age can have biological ages up to 20 years apart due to genetic predisposition and lifestyle.

What does telomerase do?

Telomerase is an enzyme that can repair / rebuild telomeres. Its activity is partly genetic — people with strong telomerase activity preserve telomeres better; those with reduced activity see faster telomere shortening, especially under oxidative stress.

Which genes are evaluated in the analysis?

Mainly telomerase and related genes (TERT, BICD1, PPARG) controlling telomere maintenance. They show how well your body can repair / preserve telomeres — a rough estimate of your genetic anti-aging capacity.

WHAT LIFESTYLE DOES

Does Mediterranean diet work for everyone?

Surprisingly no. Studies show it extends telomeres only in people with a specific gene variant — others see no measurable effect on telomere length. A clear demonstration that the same lifestyle advice does not work the same in every individual.

Which nutrients support telomeres?

Vitamin D3, Omega-3, Vitamin E, Resveratrol, and Ginkgo. When the genetic protection system is weak, these nutrients can be used deliberately to slow telomere shortening.

Can I influence free radicals?

Yes — antioxidants in food (Vitamin C, E, polyphenols from berries, tea, etc.) neutralise free radicals before they damage cells. In genetically weakened antioxidant defences, an antioxidant-rich diet is especially important.

INFLUENCE & EXPECTATIONS

How quickly will I see results from anti-aging measures?

Telomere effects are slow — months to years. Some intermediate markers (oxidative stress, inflammation, blood lipid profile) shift in weeks. The strongest impact of lifestyle on biological age compounds over decades.

What practical use does this knowledge give me?

You know whether you genetically lean toward faster or slower cell aging — and can act deliberately. With unfavourable predisposition, focus on: antioxidant-rich diet, targeted supplements (D3, Omega-3, Resveratrol), regular but not excessive exercise, good sleep, and stress reduction.